

Problem set 3

- Due date Feb 13
- Questions 14,15,16,17,18 of PS2
- Plus the following problem
- One team from teams 1,2,3,4 will be called on questions 14,15,18 of PS2 and 1,2,3 of the problem below
- One team from teams 5,6,7,8,9 will be called on questions 16,17 of PS2 and 4,5,6 of the problem below

PROBLEM

Consider the following model. There are two accumulable inputs, K , physical capital, and H , human capital. One unit of output can be either consumed, transformed into one unit of H , or transformed into one unit of K . Therefore at any date t ,

$$C_t + \dot{K}_t + \dot{H}_t = Y_t.$$

The aggregate production function is

$$Y_t = AK_t^\alpha H_t^{1-\alpha}.$$

Question 1 – Show that in any balanced growth path (BGP), one must have

$$g_C = g_K = g_H = g_Y = g,$$

where, for any variable X , $g_X = \dot{X}/X$ is its growth rate.

Assume that the economy saves a constant fraction of output in each type of capital, that is,

$$\dot{K} = s_K Y$$

and

$$\dot{H} = s_H Y,$$

with

$$s_H + s_K \leq 1.$$

Question 2 – Compute the equilibrium growth rate g , as well as the equilibrium H/K ratio h , in a balanced growth path.

Question 3 – What are the values of (s_H, s_K) that maximize g ? How do they depend on α ? Is it optimal for consumers to pick those values? Explain.

Now assume that the growth trajectory is determined in a decentralized competitive equilibrium where agents can borrow and lend at an instantaneous real rate r_t . The consumers' intertemporal utility is

$$V = \int_0^{+\infty} \frac{C_t^\beta - 1}{\beta} e^{-\rho t} dt.$$

Let

$$\sigma = \frac{1}{1 - \beta}.$$

Question 4 – Show that in a BGP, one must have

$$g = \sigma(r - \rho).$$

Question 5 – Argue that in a competitive equilibrium,

$$MPK = MPH = r.$$

Question 6 – Compute the equilibrium value of h and show that

$$g = \sigma(A\alpha^\alpha(1 - \alpha)^{1-\alpha} - \rho).$$